ABANYERONDO

Community Security Availability and Emergency Assistance System

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Course: INSY 8311 | Lecturer: Eric Maniraguha

GitHub: repository-ready package; live URL pending creation

Background and the problem

Citizen pain

A citizen away from a saved home area may not know which approved local duty contact serves the place where assistance is needed.

Operational pain

Upstream suspensions, transfers and assignments may make a directory stale. A shift schedule does not confirm an officer is available.

Evidence note: emergency-location studies in Rwanda concern EMS, not Irondo specifically.

Why this project

- A citizen should be able to search a saved home area instantly.
- Travelling to Rubavu should not overwrite a saved Kigali location.
- Only duty-verified, publicly approved contacts should be visible.
- An external status change must update public results reliably.

Main and specific objectives

Main objective

Improve citizen access to verified local community security duty contacts.

- Normalize Rwanda public administrative geography.
- Synchronize and audit external personnel events.
- Distinguish assigned, scheduled, checked-in and available.
- Provide Oracle-connected saved and alternate location search.

Research and existing solutions

Hunt et al. (2024)

Rwanda EMS location communication challenges; relevant analogy only.

Pilemalm & Mojir (2020)

Interoperable ICT for emergency coordination, not Irondo-specific.

Trek Medics Beacon

Existing responder communication/coordination platform.

Our distinguishing idea

Location preferences + public duty contacts + validated upstream events.

Real source data — validated

Open Admin Data: Rwanda administrative divisions • CC BY 4.0

5	30	416	451
Province-level	Districts	Sectors	Total real rows

Sample: RW1 Kigali City → RW12 Gasabo → RW1209 Kimironko

https://github.com/open-admin-data/rwanda-administrative-divisions

Proposed solution and innovation

External authority

Simulated upstream events: hired, transfer, suspension, dismissal.

Oracle PL/SQL

Idempotent sync, eligibility, check-in validation and auditing.

Oracle APEX

Citizen saved / new-area lookup and live approved contact directory.

Innovation: show a live transfer/suspension changing the actual Oracle-backed search result.

Planned database and relationships

Real reference geography

SOURCE_IMPORT → PROVINCE → DISTRICT → SECTOR

Foreign keys enforce the administrative hierarchy.

Synthetic operational domain

OFFICER_REFERENCE → PERSONNEL_EVENT → ASSIGNMENT → DUTY_SHIFT / DUTY_CHECKIN → DUTY_CONTACT

Phase II: ERD, 3NF proof, primary/foreign keys, historical effective periods.

PL/SQL business-rule approach

- Procedure apply_event: reject duplicate IDs and conflicting stale updates.
- Function is_available: check eligibility + current assignment + check-in + freshness.
- Cursor find_contacts: query approved contacts in selected sector.
- Trigger / audit package: record sensitive changes, block inconsistent statuses.
- Exception handling: quarantine invalid events; scheduled job flags old feeds.

Expected demonstration

- Load 451 real reference rows; verify 5 / 30 / 416 and 0 broken parents.
- Citizen saves Kigali, searches Rubavu, then still sees Kigali saved.
- Check-in permits a synthetic test duty contact to appear in its assigned sector.
- Simulated suspension hides the contact from live results immediately.
- Oracle audit report records source, time, event ID and outcome.

Limitations and future work

Current limits

Real source covers sector geography only, not cells/villages. Staff and incidents are synthetic; no official integration or authorization.

Future improvements

Obtain up-to-date licensed cell/village data, external security-system approvals, verified public duty numbers and independent geographic validation.

Safety: official emergency services remain the fallback; never represent the demo as an emergency hotline.

References (APA)

1. Hunt, M., et al. (2024). Challenges locating the scene of emergency: A qualitative study of the EMS system in Rwanda. Prehospital Emergency Care, 28(3), 501–505.

2. Pilemalm, S., & Yousefi Mojir, K. (2020). ICT enabled cross-sector collaboration in emergency response. International Journal of Emergency Management, 16(3), 249–280.

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5. Oracle. (n.d.). Coding PL/SQL subprograms and packages. docs.oracle.com